Chemical Kinetics

Introduction :-

A substance having specific properties is transformed into other substance in a chemical reaction, whose properties are different. We will want to know the following facts in a chemical reaction:

1. A chemical reaction is possible or not in given condition, this is determined by thermodynamics where the value of Gibb's free energy is obtained as negative ($-\Delta G$) then process will be possible otherwise not.

2. How many parts of reactants would be converted into products, this can not be determined by chemical equilibrium.

3. What will be the time required to complete the reaction or to reach the reaction at equilibrium.

Above facts can be determined by given conditions such as pressure, temperature, concentration, catalyst etc. and the knowledge of the mechanism of reaction is useful to understand the above facts.

The chemical reactions take place at different rates. Some reactions like acid base reactions, precipitation of $AgCl$, $BaSO_4$ etc. are completed in a fraction of second.

Some reactions like, photochemical reactions, generally take place with slow rate. Its reason is that in these reactions, actual transformations of electrons, atoms or groups take place.

Similarly all organic reactions also take place at slow rate due to molecular nature of reaction.

Besides all these, some reactions take place at very slow rate **eg:** formation of bio fuel, degradation of rocks, change in geographical layer, nuclear decay, transformation of carbon into diamond etc.

Chemical kinetics deals with the study of these changes and factors affecting changes.

"Chemical kinetics is the branch of physical chemistry which deals with the study of rate of various reactions, mechanisms and factors affecting the rate of reactions".

→ The reactions are divided in various type on the basis of their order of reactions in the study of kinetics and study of their experimental and mathematical concepts.

So, the study of kinetics represents not only reactions but also determines their conditions in which chemical changes can take place. The mechanisms of reactions are also determined from its study by which the possibilities to complete other reactions are also determined.

* Rate Of Chemical Reaction :→

All chemical reactions takes place at different rates. Some reactions take place at very fast rate while some reactions takes place at slow rate and some at very slow rate.

For example, when the aqueous solution of NaCl (sodium chloride) is mixed in aqueous solution of silver nitrate ($AgNO_3$) then white ppt. of silver chloride ($AgCl$) is formed immediately.

→ Similarly, acid-base reactions also take place at very fast rate.

Generally these reaction takes place at very fast rate. (10^{-12} to 10^{-16} seconds).

→ On the other hand, some reactions take place at very slow rate. **e.g.** rusting of iron takes place at such slow rate that this process cannot be observed by normal method.

While some reactions are also there, which takes place in measuring time and their rates can be studied in laboratories.

For example:

1. Hydrolysis of ester in the presence of sodium hydroxide →

$$CH_3COOC_2H_5 + NaOH \longrightarrow CH_3COONa + C_2H_5OH$$

2. Inversion of Canesugar →

$$\underset{\text{(Sugar)}}{C_{12}H_{22}O_{11}} + H_2O \longrightarrow \underset{\text{(Glucose)}}{C_6H_{12}O_6} + \underset{\text{(fructose)}}{C_6H_{12}O_6}$$

3. Thermal decomposition of nitrogen pentoxide →

$$2N_2O_5 \longrightarrow 4NO_2 + O_2$$

Such reactions are important as per view of chemical kinetics.

** → "The rate of reaction is defined as the change in concentration of reactants or products perunit time in a reaction".

For a hypothetical chemical reaction,

$$R \longrightarrow P$$

In this reaction, there are one mole of R reactant (molecule) and one mole for P product (molecule). The rate of reactions can be given by one of the following two methods:

(i) Rate of decrease in conc. of Reactant (R) →

$$\text{Rate of reaction} = \frac{\text{Decrease in concentration of R}}{\text{Time taken to decrease.}}$$

(ii) Rate of increase in conc. of product (P) →

$$\text{Rate of reaction} = \frac{\text{Increase in concentration of P}}{\text{Time taken to increase.}}$$

If $[R]_1$ and $[P]_1$ are concentration of R & P at time t_1 and these are $[R]_2$ and $[P]_2$ respectively at time t_2, then -

$$\Delta t = t_2 - t_1$$

$$\Delta R = [R]_2 - [R]_1$$

$$\Delta P =$$

where square brackets represent their concentration and Δt represents time internal, ΔR represents change in conc. of reac-tants and ΔP represent in conc. of products.

So,

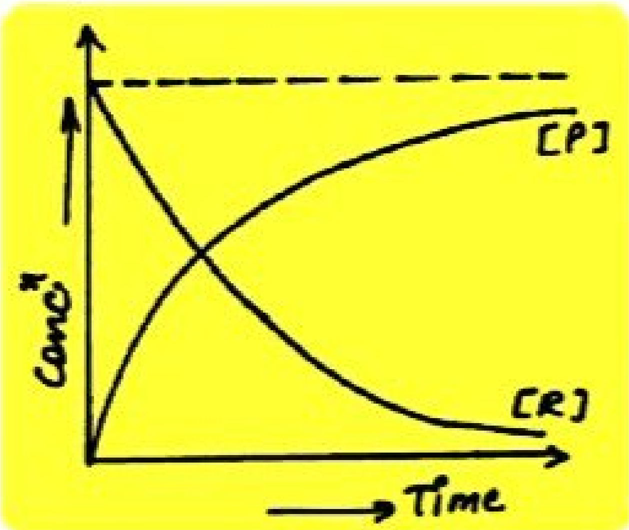
$$\text{Rate of reaction} = -\frac{\Delta R}{\Delta t}$$

or

$$\text{Rate of reaction} = +\frac{\Delta P}{\Delta t}$$

→ The concentration of reactants decreases, so it represents by (−) sign. while the concentration of products increases. So its represents by (+) sign.

Following graph is obtained on change in concentration of reactants and products with time:



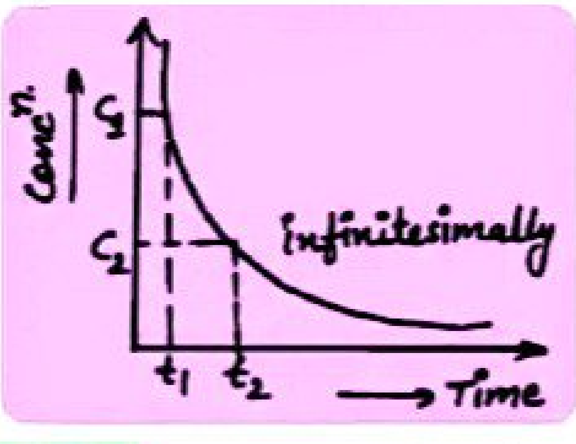
→ As reaction procede the rate of reaction decreases with time. It is clear from following graph that the rate of rx^n decreases rapidly initially and then constant at the end because rate time curve is infinitesimally. It means from theoritical point of view that infinite time is required in completion reaction.

*** Average rate** → The rate of a chemical rx^n decreases infinitesimally, so it decreases initially and constant at the end by decreasing slowly. So the average rate of a reaction is a meaningless term. But average rate can be determined for a small interval of time.

Acc. to graph, avg. rate at conc. C_1 and C_2 with time t_1 to t_2 is -

$$\boxed{\bar{V} = \pm \frac{\Delta C}{\Delta t}}$$

→ The positive and negative sign represents change in conc. with respect to products and reactants respectively



*** Instantaneous velocity** :- The actual rate of reaction at certain moment is called instantaneous Velocity.

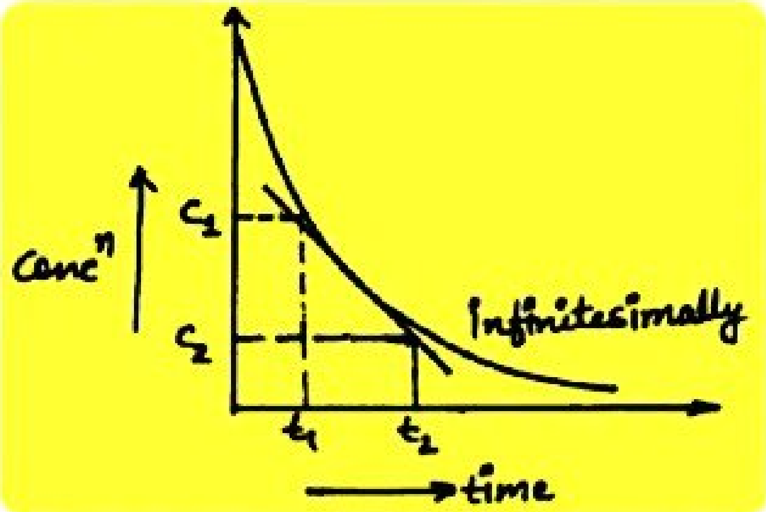
Instantaneous velocity can be determined by taking time interval (Δt) infinitesimally.

This step is important because Calculus can be used by this in study of kinetics.

If instantaneous velocity is determined at a point P then its slope is determined by drawing a contact line on it ($\tan\theta$).

It can be represented as follows in calculus:

$$\text{Instantaneous velocity}(v) = \lim_{\Delta t \to 0}\left[\pm\frac{\Delta c}{\Delta t}\right] = \pm\frac{dc}{dt}$$



Where 'dc' is infinitesimally change in concentration i.e. change in conc. and dt is infinitesimally time used in this change i.e. change in time.

Hence infinitesimally change is not possible experimently and can be obtained simply by slope of contact line, which is instantaneous velocity.

Instantaneous velocity at P =

$$\tan\theta = \frac{-\Delta c}{\Delta t} \quad \text{or} \quad \tan\theta = \frac{c_2 - c_1}{t_2 - t_1}$$

(Negative in reference to concentration of reactants)

** Relation between Rate of reaction and Stoichiometry:

The rate of complete chemical reaction is determined by the study of change in concentration of a reactant or a product with time. It is necessary for this that the chemical equation is balanced i.e. represented in stoichiometric equation.

Stoichiometry is quantitative-mass relation of chemical reaction by which balanced chemical equations are represented.

For a chemical reaction,

$$A + B \longrightarrow C + D$$

The rate of this reactions can be given by one of the following steps.

$$\text{Rate of reaction } (V) = \frac{-d[A]}{dt} = -\frac{d[B]}{dt} = +\frac{d[C]}{dt} = +\frac{d[D]}{dt}$$

→ Here, it is determined from stoichiometry of above reaction that the concentration of A and B decrease with same velocity and this velocity is equal to that velocity at which the concentration of C & D increase.

If the reaction is as follows -

$$2A \longrightarrow C + D$$

→ It is clear from stoichiometry of reaction that the rate of disappearance of A is twice the velocity of formation of C and D. So, the rate of reaction can be given as below:

$$\text{Rate of reaction } (V) = -\frac{1}{2}\frac{d[A]}{dt} = +\frac{d[C]}{dt} = +\frac{d[D]}{dt}$$

Above three steps are equivalent so rate of reaction can be determined from one of the above steps.

Similarly →

For example:

$$5Br^-(aq.) + BrO_3^-(aq.) + 6H^+(aq) \longrightarrow 3Br_2(g) + 3H_2O(l)$$

$$\text{Rate of reaction} = -\frac{1}{5}\frac{\Delta[Br^-]}{\Delta t} = -\frac{\Delta[BrO_3^-]}{\Delta t} = -\frac{1}{6}\frac{\Delta[H^+]}{\Delta t}$$

$$= \frac{1}{3}\frac{\Delta[Br_2]}{\Delta t} = \frac{1}{3}\frac{\Delta[H_2O]}{\Delta t}$$

→ The rate of reaction can be determined from any one of the above steps.

Unit of rate of Rx^n = $mol\ L^{-1} sec^{-1}$

$$\therefore \frac{\Delta C \rightarrow mol/Lt}{\Delta t \rightarrow sec.}$$

* Rate of reaction and rate constant :-

Acc. to law of mass action, the rate of reaction is directly proportional to the concentration of reactants at given temperature.

Consider the following reaction :

$$A \longrightarrow B$$

If concentration of A at given time is C_A, then acc. to law of mass action:

$$\text{Rate of reaction} = \frac{dx}{dt} \propto C_A$$

or

$$= \frac{dx}{dt} = KC_A \quad ----(i)$$

Here, $\boxed{K}$ = Proportionality constant which depends on the nature of substance, temperature etc.

It is called rate constant or rate coefficient or specific rate of reaction etc.

Consider the following reaction for defining rate constant:

$$aA + bB + cC \longrightarrow \text{Product}$$

Rate of above reaction

$$\frac{dx}{dt} \propto [A]^a [B]^b [C]^c$$

or

$$\frac{dx}{dt} = K[A]^a [B]^b [C]^c \quad ---(ii)$$

if conc^n of all reactants in above reaction are same and equal to unity,

$$\therefore \ [A]^a = [B]^b = [C]^c = 1$$

$$\therefore \ \text{Rate of reaction}, \frac{dx}{dt} = K \quad ---(iii)$$

→ Hence, specific rate of reaction or rate constant is that rate of reaction, where the conc^n of all reactants are unity.

The unit of specific rate of reaction depends on order of reaction, so the rate constant of a reaction is called specific rate of reaction.

* Units of Rate constants :

The unit of rate constant depend on the order of reaction. Consider the following reaction :

$$aA + bB + cC \longrightarrow Product$$

$$\text{Rate of reaction} = -\frac{dc}{dt} = \frac{\text{Change in concentration}}{\text{Time taken in change}}$$

$$= \text{Concentration} \times \text{time}^{-1} \quad \text{---(i)}$$

$$\text{Rate of reaction} = k[A]^a[B]^b[C]^c$$

$$\Rightarrow k[\text{conc.}]^a[\text{conc.}]^b[\text{conc.}]^c = k[\text{conc.}]^{a+b+c} \quad \text{---(ii)}$$

equation (i) & (ii) are equivalent.

$$k = [\text{conc.}]^{a+b+c} = \text{Concentration} \times \text{time}^{-1}$$

$$\Rightarrow k = \frac{\text{conc.}}{[\text{conc.}]^n}\text{time}^{-1} \quad [\text{where } n = a+b+c]$$

$$\Rightarrow k = [\text{Concentration}]^{(1-n)} \times \text{time}^{-1} \quad \text{--------(iii)}$$

→ If concentration is represented in gram mole/lt and time in second, then -

$$\left\{ \text{Rate constant}(k) = \left[\frac{\text{mole}}{\text{litre}}\right]^{1-n} \times (\text{second})^{-1} \right\} \quad \text{-----(iv)}$$

→ This is a generalized relation from which unit of rate constant for any order of reaction can be determined.

Example →

1. Unit of zero order reaction ; $n = 0$

$$k_0 = \left[\frac{\text{mole}}{\text{litre}}\right]^{1-0} \times \text{sec}^{-1}$$

The unit of k_0 is mole $\text{litre}^{-1}\,\text{sec}^{-1}$ or mole $L^{-1}s^{-1}$

2. Unit of 1^{st} order reaction ; $n=1$

$$K_1 = \left[\frac{mole}{litre}\right]^{1-1} \times sec^{-1} = \left[\frac{mole}{litre}\right]^{0} \times sec^{-1}$$
$$= second^{-1}$$

The unit of K_1 is $second^{-1}$ or s^{-1}

3. Unit of second order reaction ; $n=2$

$$K_2 = \left[\frac{Mole}{litre}\right]^{1-2} \times second^{-1} = \left[\frac{Mole}{litre}\right]^{-1} \times second^{-1}$$
$$= \frac{litre}{mole} \times second^{-1}$$

The unit of K_2 is $Litre\ mole^{-1} second^{-1}$ **or** $L\ mole^{-1} s^{-1}$

 # Order of Reaction and Molecularity:- In most of reactions, molecularity and order of reaction are same but it is not always true because both of steps are different.

★ Order of reaction: Order of a chemical reaction means – "The number of molecules or atoms of reactants participating in elementary step which determines the rate of reaction is known as order of reaction".

For example →

$$A \longrightarrow Product\ ;\ Rate(r) = K_1[A]$$

→ This is 1^{st} order reaction.

And -

$$2A \longrightarrow Product,\ Rate(r) = k_2[A]^2$$

$$A+B \longrightarrow Product,\ Rate(r) = k_2[A][B]$$

These are second order reactions.

★ Molecularity: Molecularity of a reaction means the number of molecules, atoms or ions participating in that reaction which represents stoichiometric equation.

For example →

$$A \longrightarrow \text{Product}, \text{Molecularity} = 1$$
$$2A \longrightarrow \text{Product}, \text{Molecularity} = 2$$
$$A + B \longrightarrow \text{Product}, \text{Molecularity} = 2$$
$$3A \longrightarrow \text{Product}, \text{Molecularity} = 3$$
$$2A + B \longrightarrow \text{Product}, \text{Molecularity} = 3$$
$$A + B + C \longrightarrow \text{Product}, \text{Molecularity} = 3$$

→ So, molecularity of first order is one, second order is two and molecularity of third order is three, but it is not always necessary that the molecularity of reaction is equal to order of reaction.

Molecularity is a whole number which can also be greater than order of reaction.

Besides this, same reactions may have zero order of reaction and fractional order of reaction.

The minimum value of order of chemical reaction can be zero and maximum value can also be three.

It is explained by following examples:

1. The reaction between H_2 and Cl_2 in the presence of sun light at the surface of water is **zero order reaction** while its **molecularity is two.**

$$H_2 + Cl_2 \xrightarrow[\text{surface of water}]{h\nu} 2HCl$$

2. Acidic hydrolysis of ester is **first order** (pseudo first order) **reaction** while its **molecularity is two.**

$$CH_3COOC_2H_5 + H_2O \xrightarrow{H^+} CH_3COOH + C_2H_5OH$$
$$\text{Rate} = k_1[CH_3COOC_2H_5]$$

3. Following reaction is **1st order reaction** while its **molecularity** is also **two.**

$$2N_2O_5 \longrightarrow 4NO_2 + O_2$$
$$\text{Rate} = k_1[N_2O_5]$$

→ It can be explained by its mechanism which is given as below:

$$N_2O_5 \longrightarrow N_2O_3 + O_2 \quad \text{(slow step)}$$
$$N_2O_3 \longrightarrow NO + NO_2 \quad \text{(fast step)}$$
$$N_2O_2 + NO \longrightarrow 3NO_2 \quad \text{(fast step)}$$

$$2N_2O_5 \longrightarrow 4NO_2 + O_2$$

→ The first step of this mechanism is slow step which is rate determining step. (RDS)

Molecularity of slow step is order of reaction.
So, it is clear that molecularity and order of most of the reactions are same.

Whereas it is not true for the reactions taking place in more than one step.
[All India 2014]

Difference b/w Order of reaction & Molecularity

Order of Rx^n	Molecularity of Rx^n
1. It is the sum of powers of the conc^n. in the rate law expressions.	1. It is the number of molecules of the reactants taking part in elementary reaction.
2. Order corresponds to the overall reaction.	2. Molecularity is for elementary reactions.
3. Order might involve one elementary reaction or it might involve a sequence of elementary reactions.	3. Molecularity is a part of the whole reaction.
4. It is an experimentally determined value.	4. It is a theoritical concept.
5. It can have fractional value.	5. It is always a whole number.
6. It can assume zero value	6. It can not have zero value.

7. Order of a reaction can change with the conditions such as- Pressure, Temperature, conc^n etc.	7. Molecularity is invariant for a chemical equation.

Factors affecting the rate of reaction:- The rate of reaction is affected by various factors from which some important factors are given below:

1. Concentration of the reactants.
2. Temperature of the system.
3. Nature of reactants and products.
4. Presence of catalysts
5. Surface area
6. Exposure to radiations

1. Concentration of reactants:- It is observed that the rate of reaction decreases with time. We know that initially the conc. of reactants is maximum so the rate of change in concentration is also maximum. As the concentration of reactants decreases then the rate of reaction also decreases.

It means that the rate of reaction is directly proportio--nal to the concentration of reactants.

2. Temperature of system:- Generally, the rate of all reactions approximately increases on increasing temperature. In other words, the rate of reactions also decreases on decreasing temperature. This observation is applicable on all types of exothermic and endothermic reactions.

According to normal observation, the rate of reaction mixture increases two to three times on increasing temperature upto 10°c.

Its reason is that the kinetic energy of reactant molecules increases on increasing temp. and according to collision theory, collision b/w molecules also increases which increases the number of active molecules.

As a result of this, the rate or transformation of reactant molecules into products i.e, the rate of reaction also increases.

3. Nature of reactants and products: The rate of reaction is also affected by the nature of reactants and products.

Old bonds in a chemical reaction are broken and new bonds are formed.

So, the reactivity of substance depends on breaking and formation of specific bonds.

e.g. the oxidation of nitric oxide into nitrogen dioxide takes place rapidly but the oxidation of CO into CO_2 takes place at slow rate.

$$2NO + O_2 \longrightarrow 2NO_2 \quad \text{(fast rate)}$$

$$2CO + O_2 \longrightarrow 2CO_2$$

→ The reactant and product molecules in these both reactions appear as same but rates of reaction are very different.

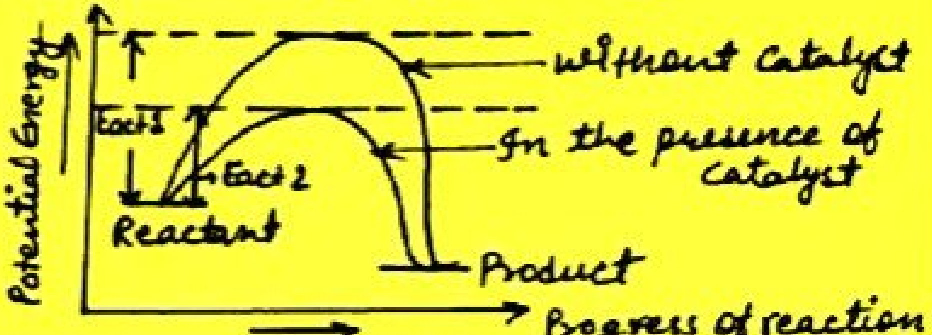
4. Effect of catalyst:- Catalyst are those external substances which increase the rate of reaction without chemical change in them.

It is seen that most of the reactions takes place relatively at fast rate in the presence of certain catalysts.

E.g.

In normal conditions, the hydrolysis of ester takes place at slow rate but it takes place rapidly in the presence of dilute acid.

$$CH_3COOC_2H_5 + H_2O \xrightarrow[\text{H}^+]{\text{(Dil. acid)}} CH_3COOH + C_2H_5$$



here -

E_{act} 1 → Activation energy without catalyst (more)

E_{act} 1 → Activation energy with / in presence (less) of catalyst

→ It is considered that the presence of catalyst decrease the activation energy of reactants which increases the rate of reaction.

5. Surface area: Greater the surface area of reactant higher is the rate of reaction. It is observed that if reactant is a solid substance then rate of reaction depends on the size of its particles.

The substance divide in such small particles, its rate of reaction is also higher because surface area of that reactant increases with small division of particles.

E.g.
A piece of wood burns slowly but it burns rapidly when cut into small pieces. The surface area of substence also increases with increase in number of pieces.
As a result of this, the rate of reaction also increases.

6. Exposure to radiation: The rate of some reactions increases rapidly in the presence of radiation (UV and Visible).

Photons of UV and visible light having high energy dissociates chemical bonds of reactants rapidly which increase the rate of reaction.

E.g. The reaction of H_2 and Cl_2 takes place at very slow rate in dark but it takes place explosively in the presence of sun light (UV and visible light).

$$H_2 + Cl_2 \xrightarrow{\text{In dark}} 2HCl \text{ (slow rate)}$$

$$H_2 + Cl_2 \xrightarrow{\text{Sun light}} 2HCl \text{ (with explosion)}$$

÷ **Zero Order Reaction :-** → Those reactions in which rate of rx^n does not depend on $conc^n$ of reactants, are called zero order reactions.

So the rate of reaction is directly proportional to zero power of $conc^n$ of reactants.

It can be explained as follows:

Rate law -

Consider the following zero order reaction,

$$A \xrightarrow{K_0} \text{Products}$$

$$\text{Rate of } rx^n = \frac{-d[A]}{dt} = \frac{dx}{dt} \propto [A]^0$$

$$\Rightarrow \frac{dx}{dt} = k_0[A]^0$$

$$\Rightarrow \frac{dx}{dt} = K_0 \qquad \text{-----(i)}$$

i.e., at any $conc^n$ of reactants the rate of reaction is constant. It is a differential equation.

Integrated equation → Equation (i) can also be written as:

$$dx = K_0 dt \qquad \text{----(ii)}$$

On integration $\int dx = K_0 \int dt$

$$x = K_0 t + C \qquad \text{---(iii)}$$

where (C = integration constt.)

When initial conditions are used for determination of Value - then,

$$t = 0$$
$$x = 0$$
$$C = 0$$

from eq^n (iii)

$$x = K_0 t$$

$$\boxed{k_0 = \frac{x}{t}} \qquad \text{----(iv)} \; (K_0 \rightarrow \text{its rate constt})$$

→ It is integrated rate equation of zero order reaction.

Unit → The generalised equation of unit is given below:

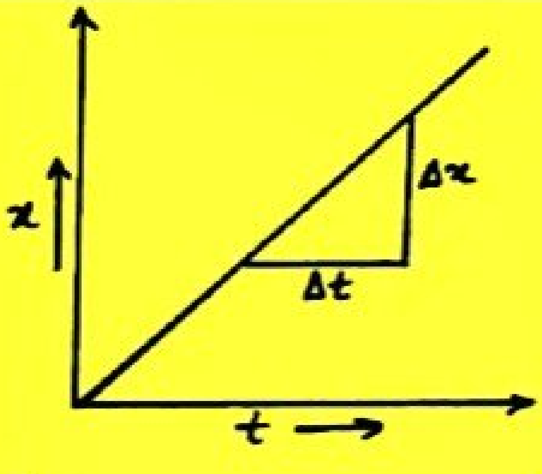
$$K_0 = \left[\frac{mole}{Litre}\right]^{1-0} \times sec^{-1}$$

The unit of K_0 is mole $litre^{-1} sec^{-1}$ or $mol\,L^{-1} s^{-1}$.

Graphical Representation →

The rate equation $x = K_0 t$ in zero order reaction is integrated rate equation.

It is an equation of straight line ($y = mx$) whose slope ($\tan\theta$) is equivalent to rate constant (K_0).



$$\left\{ slope(\tan\theta) = \frac{\Delta x}{\Delta t} \times K_0 \right\}$$

Half life :→ ($t_{1/2}$)

The time required to decompose half of the conc^n of reactants in a chemical reaction is called its life.

If the initial conc^n of reactents is 'a' then conc^n at $t_{1/2}$ will be $a/2$, so from eq^n (iv) -

$$t_{1/2} = \frac{a/2}{K_0}$$

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$$\Rightarrow \boxed{t_{1/2} = \frac{a}{2K_0}} \quad \text{------- (V)}$$

→ It is clear from equation (v) that the half life of zero order reaction is directly proportional to the initial concⁿ.

$$t_{1/2} \propto a$$

Some examples of zero order reactions are given below:

1. Reaction of H_2 and Cl_2 at the surface of water at constant pressure in the presence of light.

$$H_2 + 2Cl \xrightarrow{h\nu} 2HCl$$

2. Dissociation of phosphene at the surface of molybdenum (Mo) at high pressure.

$$2PH_3 \xrightarrow[\text{(High pressure)}]{Mo} 2P + 3H_2$$

3. Dissociation of NH_3 at the surface of tungsten.

$$2NH_3 \xrightarrow[\text{(High pressure)}]{W} N_2 + 3H_2$$

÷ **<u>First Order Reactions</u>:-** Those reactions in which the rate or reaction is directly proportional to the concⁿ. of one reactant only i.e., the rate is proportional to first power of concentration of reactant, are called first order reactions.

<u>Rate law</u>: Consider the following 1^{st} order reaction

	A ⟶	Product
Initial conc:-	a	0
Conc. at time (t):-	$(a-x)$	x
	(mole/litre)	(mole/litre)

→ According to above equation, initial concⁿ of A is mole/litre at $t=0$ which reduces to $(a-x)$ mole/litre after time 't', while x mole/litre product is obtained.

According to law of mass action,

$$\left[\frac{dx}{dt}\right] = -\frac{d(a-x)}{dt} = \frac{-d[A]}{dt} \propto (a-x) \quad ---- (i)$$

So,
$$\frac{dx}{dt} = k_1(a-x) \quad ---- (ii)$$

where, K_1 = rate constant of 1st order reaction and equation (ii) is the differential equation of rate of 1st order reaction.

Integrated rate equation:

On integrating equation (ii),

$$\int\frac{dx}{(a-x)} = K\int dt$$

$$\left[\text{Formula} \int\frac{dx}{x} = \ln x\right]$$

$$-\ln(a-x) = k_1t + c \quad ---- (iii)$$

Initially at $t=0, x=0$, then

$$-\ln a = c \quad ---- (iv)$$

From equation (iii) and (iv)

$$-\ln(a-x) = k_1t - \ln a$$

$$\Rightarrow \ln a - \ln(a-x) = k_1t$$

$$\ln\frac{a}{a-x} = k_1t \quad ---- (v)$$

$$\Rightarrow k_1t = 2.303 \log\frac{a}{a-x}$$

$$\Rightarrow \boxed{k_1 = \frac{2.303}{t}\log\frac{a}{(a-x)}} \quad ---- (vi)$$

→ This is integrated rate equation of 1st order reaction.

* Interval Equation →

If the initial concentration of reactant A is not known then interval equation can be used to determine rate constant.

In this formula, initial concentration is removed from rate equation.

Suppose, x_1 and x_2 amounts of reactants have decomposed at two different times t_1 and t_2 respectively.

Equation (vi) can be written as below:

$$\Rightarrow t_1 = \frac{2.303}{k_1} \log \frac{a}{(a-x_1)} \quad \text{------ (vii)}$$

$$\Rightarrow t_2 = \frac{2.303}{k_1} \log \frac{a}{(a-x_2)} \quad \text{----- (viii)}$$

→ By substracting eq. (vii) from eq. (viii) -

$$(t_2 - t_1) = \frac{2.303}{k_1}\left[\log \frac{a}{(a-x_2)} - \log \frac{a}{(a-x_1)}\right]$$

$$\Rightarrow (t_2 - t_1) = \frac{2.303}{k_1} \log \frac{a-x_1}{a-x_2} \quad \text{--- (ix)}$$

* Graphical Representation :→

we can draw various types of graphs for 1^{st} order reaction.

Mathematical equation of first order reaction is:

$$t = \frac{2.303}{k_1} \log \frac{a}{a-x}$$

$$\Rightarrow \log \frac{a}{a-x} = \frac{k_1}{2.303} t \quad \text{------ (x)}$$

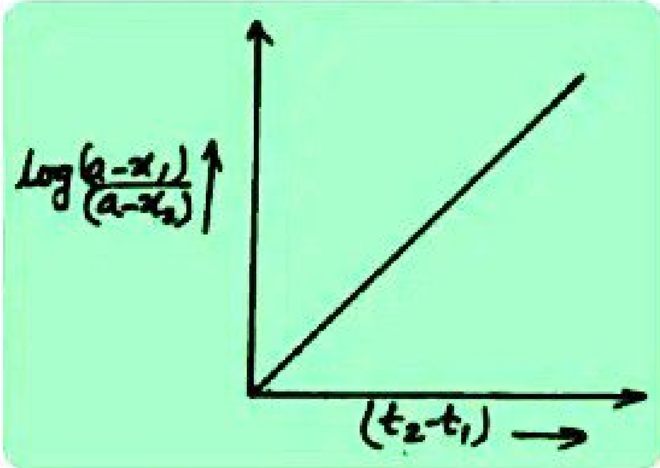
if a graph is plotted b/w $\log a/a\text{-}x$ and t then according to figure, a straight line ($y = mx$) passing through main point is obtained, whose slope is:

$$\tan\theta = \frac{k_1}{2.303} \quad \text{---- (xi)}$$

→ Graph can also be plotted from interval formula -

$$\log \frac{a-x_1}{a-x_2} = \frac{k_1}{2.303}(t_2-t_1) \quad \text{--- (xiii)}$$

its slope is → $\tan\theta = k/2.303$



★ **Half life** → Initially, the conc. of reactant is 'a' mole/lt and time required to half its conc. a/2 is called its half-life so, from mathematical equation,

$$\Rightarrow t_{1/2} = \frac{2.303}{k_1} \log \frac{a}{a-a/2}$$

$$\Rightarrow t_{1/2} = \frac{2.303}{k_1} \log 2$$

$$\Rightarrow t_{1/2} = \frac{2.303 \times .3010}{k_1} \quad [\because \log 2 = .3010]$$

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So, $$\boxed{t_{1/2} = \frac{0.693}{k_1}} \quad \text{---- (xiv)}$$

→ This represents the half-life of first order reaction, It is clear from half-life equation that the half-life of 1st order reaction depends on the initial conc. of reactants.

★ **Unit of k** → $\left[\frac{\text{mole}}{\text{Litre}}\right]^{1-n} \times \text{second}^{-1}$ [n = 1]

$\Rightarrow$ $\underline{\text{second}^{-1}}$ or $\underline{\text{sec}^{-1}}$

The examples of first order reaction are as follows :

1. On heating aqueous solution of ammonium nitrite -

$$NH_4NO_2 \xrightarrow{\Delta} 2H_2O + N_2$$

2. Thermal decomposition of azo isopropane -

$$(CH_3)_2CH-N=N-CH-(CH_3)_2 \xrightarrow{\Delta} N_2 + C_6H_4$$

3. All radioactive decays are 1^{st} order reactions.

★ Pseudo First order reaction :→

The order of a reaction is sometimes altered by conditions. There are many reaction which obey 1^{st} order rate law, although they are higher order reaction.

E.g. The hydrolysis of ethyl acetate which is a chemical reaction b/w ethyl acetate and water.

In reality, it is a 2^{nd} order reaction and concentration of both ethyl acetate and water affect the rate of the reactions. But water is taken in large excess for hydrolysis, therefore - conc. of water is not altered much during the reaction.

→ Thus the rate of reaction is affected by conc. of ethyl acetate only. e.g. during the hydrolysis of 0.01 mol of ethyl acetate with 10 mol of water, amounts of the reactants and products at the beginning (t = 0) and completion (t) of the reaction are given as under.

	$CH_3COOC_2H_5$ +	H_2O $\xrightarrow{H^+}$	CH_3COOH +	C_2H_5OH
t=0	0.01 mol	10 mol	0 mol	0 mol
t=t	0 mol	9.9 mol	0.01 mol	0.01 mol

→ The conc. of water does not get altered much during the course of the reaction, so the rx^n behaves as first order rx^n. Such rx^n are called **pseudo 1^{st} order rx^n**.

→ Inversion of cane sugar is another pseudo first order reactio.

$$\underset{\text{(Cane sugar)}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{H^+} \underset{\text{Glucose}}{C_6H_{12}O_6} + \underset{\text{Fructose}}{C_6H_{12}O_6}$$

$$\{Rate = k[C_{12}H_{22}O_{11}]\}$$

Dependence of reaction rate on Temperature :→

The rate of rx^n is highly affected by temperature, the rate of approximately all reactions ↑se with ↑sing temperature.

Beside it, the rate of reactions ↓se with ↓se in Temp.

→ This observation is observed in all reactions, whether rx^n is exothermic or endothermic.

E.g. The rate constant of decomposition of N_2O_5 at 273 K is $7.87 \times 10^{-7}\ sec^{-1}$ which ↑se to $3.56 \times 10^{-5}\ sec^{-1}$ at 298K temperature.

Hence, the rate of this reaction increases (↑ses) to 45 times on ↑sing 25°c temperature.

→ According to a normal observation, the rate of chemical reactions increases 2 to 3 times on ↑sing 10°c temperature. It is called "temperature coefficient".

So,

$$\left\{ \text{Temp coefficient} = \frac{\text{Rate constant at } (T+10)^\circ c}{\text{Rate constant at } T^\circ c} \right\}$$

$$= 2 \text{ to } 3$$

→ According to collision theory, the rate of a chemical rx^n depends on -

1 → Collision frequency (z)

2. → Fraction of effective collisions (f)

→ The ↑se in rate on ↑sing temp. can be explained by above two factors or one of the factors.

(1) By increasing Collision Frequency: The average kinetic energy of reactant molecules increases with ↑se in temperature, which also ↑ses the number of collisions take place per second in per unit volume.

The avg. kinetic energy of molecules is directly proportional to absolute temperature.

According to an observation, the K.E. of molecules ↑ses to approx. 3% on ↑sing temp. from 300K to 310K.

→ It means that the rate of reaction ↑ses on ↑sing collision frequency with ↑se in temperature.

But it is not true because if it is true then the rate should be ↑sed to 3% on ↑sing 10°c tempr.

In fact, the rate ↑ses 2 or 3 times on ↑sing 10°c tempr which is 200 to 300%, so, the reason of ↑se in rate with ↑se in tempr is not only ↑se in collision frequency.

(2) Increase in effective collisions:→ Acc. to effective collision, only some fraction of all collisions is capable in chemical change, which are called effective-collisions.

Effective collisions are those collisions from which the released energy is equal to threshold energy or more than threshold energy.

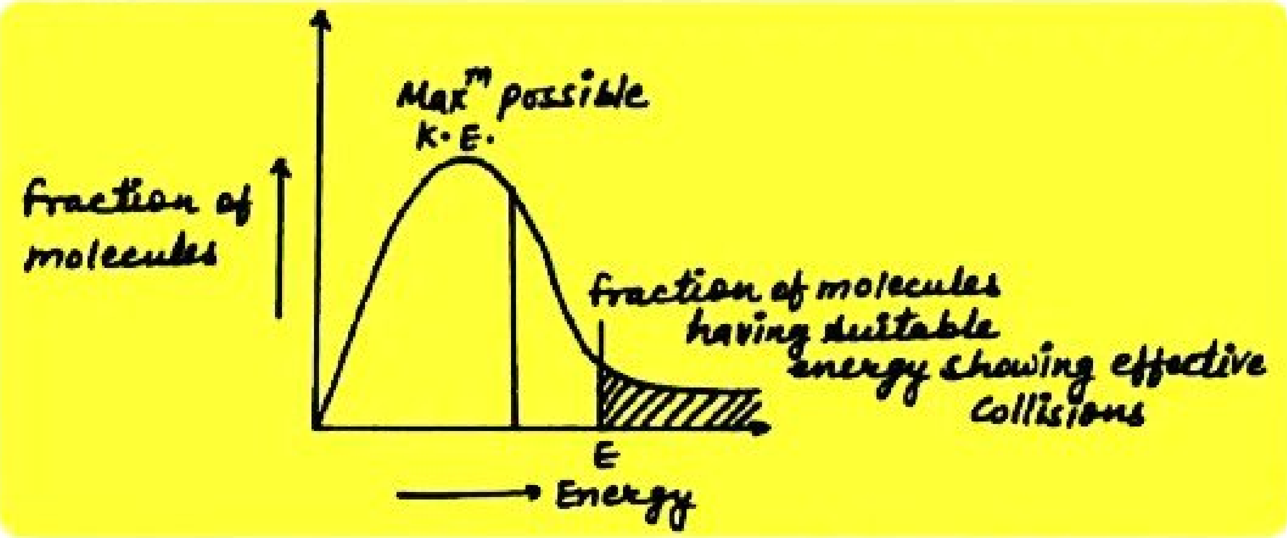
If energy released from collision is less than threshold energy then these collisions are not capable in chemical change. so, these are called ineffective collisions.

Acc. to Maxwell-Boltzmann, energy distribution of molecules can be represented on the basis of statistics.

The energies of all reactant molecules are not similar. Collision take place in molecules with each other by which these molecules transfer energy to each other.

If a graph is plotted b/w energy of molecules and fraction of molecules having energy, then graph is obtained.

This graph is called **"Maxwell-Boltzmann energy distribution graph"**.



→ It is clear from the graph that the fraction of molecules having less energy and high energy is relatively less.

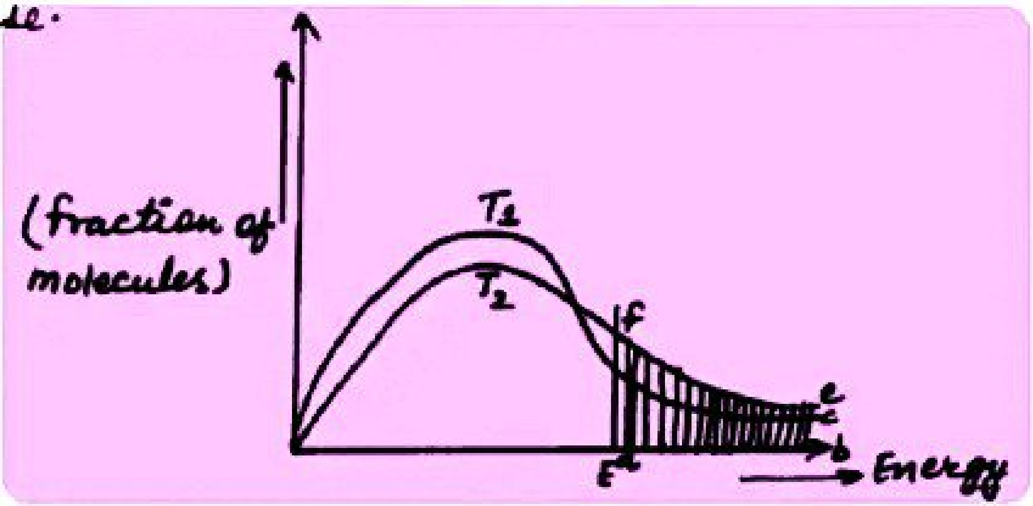
The fraction of most of the molecules has intermediate K·E· which is represented by maximum in graph and this is called "maximum possible kinetic energy".

→ The energy E represented in graph is that minimum energy i.e., threshold energy which is obtained from effective collisions and is capable to convert the reactants into products.

The molecules having energy E or higher energy are converted into products which is a small fraction of distribution of complete energy.

It can be seen that the lesser the value of threshold energy E, the number of effective collisions will be higher, so the rate of reaction will be higher.

Whereas with the ↑se of value of E the number of effective collisions will ↓se and the rate of reaction will also decrease.



→ Now, Consider the number of effective collisions on ↑sing temp. The distribution of energy of molecules at tempʳ T_1 and T_2 is represented in previous graph.

$$\text{where } T_2 = T_1 + 10^\circ c$$

→ It is clear from graph that the graph is stretched to right side on ↑sing temperature and becomes more flat.

It is determined that energy distribution in molecules ↑ses on ↑sing tempᵀ and fraction of molecules having threshold energy (E) increases, abcd fraction shows effective collisions at tempʳ. T_1 which ↑ses to abef at temperature T_2.

→ This fraction is approx. twice on ↑sing tempʳ of 10°c. so, it is clear that "the main reason of ↑se in rate of reaction on ↑sing tempʳ. is ↑se in number of effective collisions."

* Arrhenius Equation and Calculation of Activation Energy :-

Arrhenius proposed following equation by relating temperature and activation energy with rate constant.

$$K = Ae^{Ea/RT} \quad \text{---- (i)}$$

→ This relationship is called Arrhenius equation where A is arr--henius constant or Frequency constant.

This constant represents bi-molecular collisions per second in per litre volume. Here R is gas constant and E_a represents activa--tion energy. T is absolute temperature. By taking logarithm of eq. (i)

$$\ln k = \ln A - \frac{E_a}{RT} \quad \text{-----(ii)}$$

By connecting exponential logarithm is to logarithm.
($\ln = 2.303 \log$)

$$\log k = \log A - \frac{E_a}{2.303RT}$$

$$\log k = \frac{-E_a}{2.303R} \cdot \frac{1}{T} + \log A \quad \text{----(iii)}$$

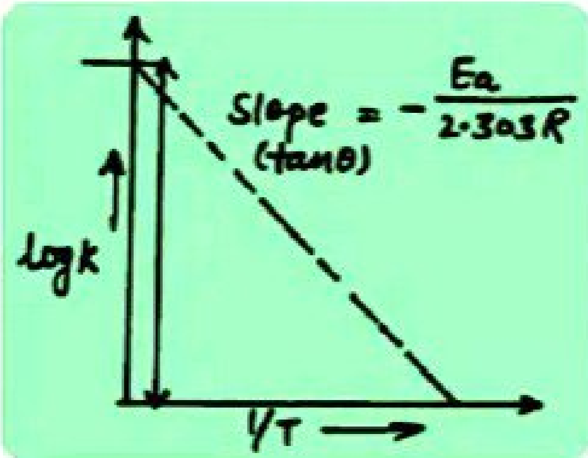
→ This is equation of straight line of $y = mx + c$ type so, a straight line having opposite slope is obtained on plotting the graph b/w $\log k$ and $1/T$.

It is also clear from equation that the value of rate constant (k) ↓ses with increase in activation energy (E_a). Activation energy can be calculated by slope of straight line having opposite slope obtained in graph.

So,

$$\text{slope} (\tan\theta) = -\frac{E_a}{2.303R} \quad \text{-----(iv)}$$

Activation energy $E_a = -2.303R \times \text{slope}$.



→ It can also be given in the form of interval if the rate constants are K_1 and k_2 at T_1 and T_2 temperature respectively, then,

$$\log K_1 = -\frac{E_a}{2.303R}\frac{1}{T_1} + \log A \quad \text{-----(v)}$$

$$\log K_2 = -\frac{E_a}{2.303R}\frac{1}{T_2} + \log A \quad \text{-----(vi)}$$

$$\Rightarrow \quad \log\left[\frac{K_2}{K_1}\right] = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$$

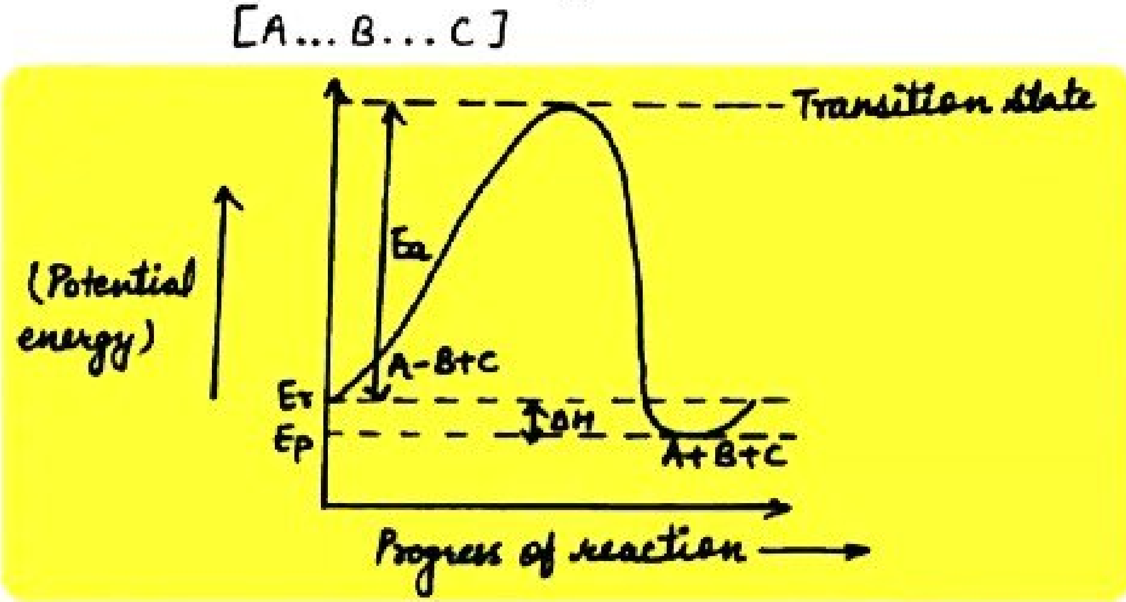
* **Energy of activation and transition state:** The minimum extra energy given to reactant molecules which is required to complete the chemical reaction is called activation energy. It is equal to difference between threshold energy and energy of reactants.

Activation energy (E_a) = Threshold energy − Avg. K.E. of reactants

$$E_a = E_{(threshold)} - E_{(reactant)}$$

→ Activation energy is specific for each reaction which depends on temp^r and fraction of total effective collisions is also determined by activation energy.

If activation energy is less in a reaction then rate of reaction ↑ses with ↑se in effective collisions.



Rate of reaction ∝ $\frac{1}{\text{Activation energy}}$

E_t = Threshold energy
E_R = Energy of reactants
E_a = Activation energy
ΔH = Enthalpy of reaction.

→ Reaction is as follows in above energy diagram −

$$\underset{\text{Reactant}}{A-B+C} \longrightarrow \underset{\text{Transition state}}{[A\ldots B\ldots C]} \longrightarrow \underset{\text{Product}}{A+B-C}$$

So, reactants get unstable state of highest energy by gaining activation energy, which is called transition state.

Reactants are converted into products by getting transition state. The energy obtained from ineffective collisions is always lesser than activation energy due to which transition state is not obtained.

So, Products are not obtained by such Collisions.

** Collision Theory :

We know that fundamental requirement for the Completion of reaction is that there is effective collision b/w reactive Species (molecules, atoms or Ions). This the basis of collision theory of reactions.

This theory was developed by **Max Troutz** and **William Lewis** in 1916-18 which is based on famous kinetic theory of gases.

→ According to this theory, reactants are considered as solid spheres and reaction take place by collision of these spheres with each other. "The number of collisions takes place per second in unit volume of reaction mixture is called collision frequency (z). Collision frequency is generally very high".

The collision frequency in a gaseous system may be 10^{25} to 10^{26} at normal temperature and pressure so, if reaction takes place in each collision then reactions will be completed in limited time but generally reaction takes place slowly.

Hence, each collision taking place b/w reactants is not capable for chemical change at given temperature.

The collisions which are actually capable to convert the reactants into products are called "effective collisions".

→ Two barriers are produced mainly in a chemical reaction:

1. Energy barrier
2. Orientational barrier.

1. Energy barrier :- There should be enough energy for effective collisions b/w reactive species (molecules, atoms or ions) by which chemical bonds of reactants can broken.

→ The minimum energy required for collisions of molecules for this work is called "threshold energy".

So only those colliding reactant molecules are converted into products which have threshold energy or more than this energy.

Threshold energy is that energy barrier with out crossing this barrier, reactants can not be converted into products.

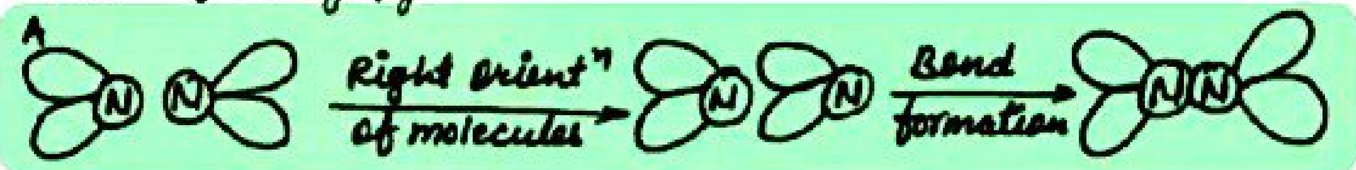
2. Orientational barrier :→ Reactant molecules collide with each other by orientation in right direction then new bonds are formed by breaking old bonds.

Collisions b/w molecules in right direction are effective collisions.

E.g. Consider the following reaction –

$$NO_{2}(g) + NO_{2}(g) \longrightarrow N_{2}O_{4}(g)$$

→ Reactants that collide by orientation in right direction are converted into products during the reaction which is represented in following figures:



$$NO_2 + NO_2 \xrightarrow[\text{Collision}]{\text{Effective}} N_2O_4$$

→ On the other hand when reactants that collide by orientation in wrong direction are not converted into products, which are represented in following figures:

$$NO_2 + NO_2 \xrightarrow[\text{Collision}]{\text{Ineffective}} NO_2 + NO_2$$

→ So, if colliding reactants have not enough threshold energy or collisions do not takes place in right direction then products are not obtained from such collisions.

In normal conditions, the faction of effective collisions is zero to one in most of the reactions.

So the rate of reaction is directly proportional to following factors:

1. Number of collisions per second in unit volume of rxⁿ mixture. i.e., collision frequency (z).

2. Fraction (f) of effective collisions i.e., orientation in right direction and fraction of reactants having enough minimum energy (threshold energy)

So,

$$\boxed{\text{Rate} = \frac{dz}{dt} = f \times z}$$